

claude-windows / 45ea3a91-654d-4972-9cd3-65f35db54caf

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\45ea3a91-654d-4972-9cd3-65f35db54caf\subagents\workflows\wf_2a64bc88-2a8\agent-a14f19923fa9118d5.jsonl`  
- SHA-256: `4fda8d7adc60971342aaa79268e1f7b1904553785857c7122da791486c8f95cd`  
- Source modified: `2026-06-10T06:03:13+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_2a64bc88-2a8`  
- session_id: `45ea3a91-654d-4972-9cd3-65f35db54caf`
```

Transcript

1. user (2026-06-10T05:59:28.003Z)

CONTEXT ? two sibling repos on a Windows machine:

- INDEXER: C:/Users/avidu/Projects/badminton-highlight-indexer ? local AI badminton(->racket-sports) highlight indexer + rally detector. FastAPI + SQLite + ffmpeg + GPU CV via WSL; Gemini = paid cloud AI. docs/ tree is rich. Current branch docs/next-steps-multisport (master = main branch).

- COLLECTOR: C:/Users/avidu/Projects/sports-data-collector ? golden-label acquisition/curation/ingestion CLI (system-of-record candidate for training corpus).

Project state (June 2026): scaffolding complete; owned-model roadmap agreed (docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md) but NO platform/owned-model implementation started; next committed platform slice = async job model. Licensing guardrail: MIT/Apache-2.0/BSD only for code+data+weights; paid LLMs are inference-only, never training-data sources. ENVIRONMENT RULES: this dev environment has NO GPU/torch/cv2 ? do NOT attempt to run the video pipeline or import torch/cv2 modules. Test suites are offline/fully-mocked and SAFE to run. You are READ-ONLY: never edit/create/delete repo files; running tests and git read commands is allowed.

QUALITY BAR: cite file paths (and line numbers where possible) for every finding. Report what IS in the code/docs, not what you assume. Be skeptical and concrete. Your final structured output is consumed programmatically by an orchestrator ? return raw data, not prose for a human.

ROLE: skeptical reviewer of a PMF research report. Below is the report. Check its 4-5 most load-bearing market claims with quick web searches (load WebSearch via ToolSearch if needed); flag any unsupported, stale, or cherry-picked claims; flag motivated reasoning in the verdict; name anything important it MISSED. Return a concise plain-text critique (bullet list).

REPORT: {"summary":"Internal thesis (docs/PMF_MARKET_RESEARCH.md, docs/COMMERCIALIZATION.md §6, docs/PLATFORM_ARCHITECTURE.md): beachhead = India (Bangalore) badminton coaching academies; wedge = software-only BYO-camera upload of full static-camera match video -> auto rally segmentation + highlight reels, evolving to conversational video-QA 'assistant coach' and multisport racquet; pricing anchors B2C \$60-180/yr, B2B academy \$300-1,500/yr (PMF_MARKET_RESEARCH.md:46-48); cost model Gemini 2.5 Flash ~\$0.05 input per 10-min clip (COMMERCIALIZATION.md:122-124); claimed whitespace 'incumbents ignore badminton, only direct competitor BadPro+ is early' (PMF_MARKET_RESEARCH.md:160-164). External research VERIFIES the pricing/cost benchmarks (SwingVision \$179.99/yr, PB Vision \$99/yr, Trace \$180-300/yr, Gemini Flash \$0.30/1M input live) but FALSIFIES the India-whitespace claim: at least four funded, celebrity-distributed Indian/Bangalore badminton AI players exist (Machaxi - \$1.5M from Rainmatter + Prakash Padukone; VISIST.AI - 70+ Bengaluru academies, Padukone School partnership; Game Theory - Gopichand-backed smart courts with auto highlights; Stupa Sports Analytics - \$3.3M, BWF-approved badminton IRS), none of which appear in the internal competitive table. The residual defensible wedge is narrower than the docs claim: indexing EXISTING messy multi-court academy footage (a problem the market only solves with per-court installed hardware), not 'badminton AI for India' generically.", "market_landscape":[{"claim":"SwingVision Pro is \$14.99/mo or \$179.99/yr, tennis + pickleball only; no badminton support today. Confirms internal pricing benchmark exactly (PMF_MARKET_RESEARCH.md:139).", "source":"https://swing.vision/subscribe", "relevance":"Closest analog; validates the \$180/yr consumer ceiling for racket-sport AI

software."},{ "claim": "SwingVision has ~20,000 paying subscribers, claims \$4M+ ARR; FY2024 revenue \$2.75M, up only ~12% YoY from \$2.46M (2023); raised \$10M+ total." , "source": "https://kingscrowd.com/swingvision-on-wefunder-2025/" , "relevance": "WTP exists but the closest analog grows slowly ? tempers B2C upside assumptions; also implies paid conversion ~10% of its claimed 200k monthly users." } , { "claim": "SwingVision raised \$6M explicitly to expand into pickleball and padel; expansion to other racket sports incl. badminton described as applicable/on roadmap." , "source": "https://www.sportspro.com/news/swingvision-ai-tennis-app-pickleball-padel/" , "relevance": "Internal risk #5 (competitive response, PMF_MARKET_RESEARCH.md:201-203) is live, not hypothetical." } , { "claim": "BadPro+ (founded by an Olympic badminton athlete + Microsoft engineer) sells pay-per-match AI badminton analysis with no subscription, plus federation and per-call API tiers." , "source": "https://www.badproplusapp.com/" , "relevance": "Confirms internal read (PMF_MARKET_RESEARCH.md:140): real badminton AI demand signal but unproven subscription WTP." } , { "claim": "Machaxi (Bengaluru) raised \$1.5M in 2025 led by Rainmatter (Zerodha) with Prakash Padukone participating; AI-powered badminton coaching centers; plans 1,000+ centres with Padukone School of Badminton across India." , "source": "https://india.entrepreneur.com/news-and-trends/machaxi-secures-15-million-to-expand-ai-powered-badminton/493070" , "relevance": "Funded, celebrity-distributed badminton AI in the exact beachhead city ? absent from internal competitive table." } , { "claim": "VISIST.AI (Bengaluru, IISc-incubated) does AI video analysis of badminton technique/skill assessment; 70+ Bengaluru academies use it; partnered with Padukone School of Badminton." , "source": "https://theprint.in/india/visist-ai-partners-with-padukone-school-of-badminton-to-digitise-athlete-skill-assessments/2716417/" , "relevance": "Directly contradicts 'incumbents ignore badminton' for the #1 beachhead; owns academy relationships the product would need. Note: skill assessment, not full-match rally segmentation/highlights." } , { "claim": "Game Theory (Bengaluru, Gopichand/Bopanna/Sharath Kamal investors, Rainmatter-backed) runs smart courts across badminton/pickleball/padel/TT with built-in recording and 'Instagram-ready video highlights' delivered in-app after booked games." , "source": "https://yourstory.com/2024/02/computer-vision-game-theory-sports-tech-startup-nithin-kamath-rainmatter" , "relevance": "Amateur highlight delivery already happening in Bangalore ? bundled with court booking, implying highlights alone may be a feature, not a product." } , { "claim": "Stupa Sports Analytics (India) raised INR 28 Cr (~\$3.3M) pre-Series A; started in table tennis, expanded to badminton; its AI instant-review system is BWF-approved; serves 15+ federations." , "source": "https://inc42.com/buzz/stupa-sports-analytics-nets-inr-28-cr-to-offer-ai-solutions-to-sports-federations/" , "relevance": "Indian racket-sports CV company with federation distribution; also validates the repo's table-tennis transfer finding (NEXT_STEPS.md:20-23) as a real market." } , { "claim": "PB Vision (pickleball AI) charges \$99/yr (~100 min analysis/mo) or \$19.99/mo; its partner API is priced at \$8 per hour of HD video processed." , "source": "https://pb.vision/" , "relevance": "Sits at the BOTTOM of the internal B2C band (\$60-180/yr); the \$8/hr API price is a direct external benchmark for the planned hosted per-video API." } , { "claim": "Veo Cam 3 ~\$1,533 hardware + Team \$49/mo / Club \$149/mo subscriptions; Trace is \$180/yr (Basic) - \$300/yr (Pro) with free leased camera." , "source": "https://www.veo.com/pricing and https://traceup.com/pricing" , "relevance": "Verifies internal hardware-bundled benchmarks (PMF_MARKET_RESEARCH.md:134-136) are current." } , { "claim": "Baseline Vision sells a \$2,000-2,300 portable AI tennis camera (line calling + analytics)." , "source": "https://www.baselinevision.com/product" , "relevance": "Hardware price point amateurs do pay in tennis; reinforces that capture hardware, not software, is where incumbents monetize." } , { "claim": "Padel already has multiple AI-camera highlight vendors: Wingfield (tennis/padel/pickleball, used by Rafael Nadal Academy), Clutch ('AI camera for padel - automated highlights'), Padelytics, PlaySight padel SmartCourt." , "source": "https://www.wingfield.io/en and https://www.clutchapp.io/" , "relevance": "Racket-sport auto-highlights is a crowded category in padel; differentiation must be sport (badminton) + footage type (BYO multi-court), not the highlight concept." } , { "claim": "WSC Sports (550+ orgs: NBA, NHL, LALIGA, DAZN) and Magnifi/Reely sell enterprise broadcast highlight automation with opaque custom pricing; Reely targets esports; none offer amateur self-serve for court sports." , "source": "https://wsc-sports.com/ and https://magnifi.ai/products/reely" , "relevance": "Confirms internal claim that broadcast highlight giants do not serve the amateur static-camera segment." } , { "claim": "The 'multi-court' problem is solved in-market only by per-court installed hardware (PlaySight SmartCourt ? up to 24 cameras; Hudl Focus; Pixellot); no software-only product found that handles adjacent-court contamination in a single static camera view." , "source": "https://playsight.com/ and https://www.hudl.com/products/focus" , "relevance": "The repo's multi-court robustness finding (NEXT_STEPS.md:23-26) targets a genuinely unsolved software gap ? but the market's revealed alternative is hardware-per-court." } , { "claim": "Gemini 2.5 Pro 'rivals specialized fine-tuned models' on video benchmarks incl. QVHighlights moment retrieval and YouCook2 dense captioning,

processing at 1fps / max 256

frames.", "source": "https://developers.googleblog.com/en/gemini-2-5-video-understanding/", "relevance": "Coarse highlight extraction from long video is commoditizing ? a thin Gemini wrapper can ship the v0 product."}, {"claim": "2025 video-LLM temporal-grounding literature (TimeExpert, Time-R1, TimeScope, VTimeLLM line) consistently reports that VLMs 'lack the ability to identify boundaries' and struggle with precise moment boundaries; the F3Set benchmark (fast/frequent/fine-grained racket-sport events, 1,000+ event types with precise timestamps) finds 'substantial challenges for existing techniques'."}, {"source": "https://arxiv.org/pdf/2508.01699 and https://arxiv.org/abs/2504.08222", "relevance": "Supports the internal head-before-VLM call (OWNED_MODEL_IMPLEMENTATION_PLAN.md:46-52): precise rally boundaries remain un-commoditized ? for now."}, {"claim": "Live Gemini API pricing: 2.5 Flash \$0.30/1M input (video) / \$2.50 output; Flash-Lite \$0.10/1M; Gemini 3 Flash preview \$0.50/1M input; batch tiers are half price."}, {"source": "https://ai.google.dev/gemini-api/docs/pricing", "relevance": "Verifies COMMERCIALIZATION.md:122-124 cost assumptions (~\$0.05 input per 10-min clip ? \$0.30/hr) ? COGS per video stays trivially low vs PB Vision's \$8/hr market price."}, {"claim": "India sports-tech: ~2,120 companies, 197 funded, \$1.97B total raised; sector projected to pass \$3.5B by 2027; Khelo India has 306 accredited academies and 1,045 grassroots Khelo India Centres."}, {"source": "https://tracxn.com/d/explore/sports-tech-startups-in-india/_4UiJ6xHvc6LY7m5rZxESDNUf11gnBudqw-M12IEDuR0 and https://kheloindia.gov.in/uploads/Khelo_India_Accrediated_Academies.pdf", "relevance": "Macro tailwind is real, but funding flows to venue/coaching plays (Machaxi, Game Theory), not standalone analysis SaaS."}, {"claim": "Bengaluru badminton academy group coaching runs roughly INR 1,500-5,000/month per student."}, {"source": "https://www.decathlon.in/blog/Top-8-Badminton-Academies-in-Bangalore", "relevance": "WTP context for the \$300-1,500/yr (INR ~25k-125k) academy price: equal to 1-3 students' monthly fees ? plausible but must displace real budget."}], "critique_of_internal_assumptions": ["FALSIFIED ? 'incumbents ignore badminton; the only direct badminton AI (BadPro+) is early' (PMF_MARKET_RESEARCH.md:163-164, COMMERCIALIZATION.md:141-144). At least four funded India/badminton AI players exist and are concentrated in the exact beachhead city: Machaxi (\$1.5M, Rainmatter + Prakash Padukone), VISIST.AI (70+ Bengaluru academies, Padukone School partnership), Game Theory (Gopichand-backed smart courts with auto-highlights), Stupa (BWF-approved badminton IRS, \$3.3M). The 2026-06-07 research snapshot missed the entire Indian competitive set despite recommending India as beachhead #1."}, {"claim": "SURVIVES NARROWLY ? the specific product gap. None of the Indian players does the repo's exact wedge: automated rally segmentation + highlight compilation from a user's OWN full-match static-camera footage as a standalone software product (VISIST = technique/skill assessment; Machaxi = coaching venues; Game Theory = highlights only on THEIR instrumented courts; Stupa = federation/broadcast tooling; BadPro+ = per-match analysis app). The whitespace is real but much narrower than 'badminton whitespace'."}, {"claim": "WEAKENED ? 'sell to the academy' B2B motion (PMF_MARKET_RESEARCH.md:165-166). Indian competitors do not sell SaaS to academies; they own venues, run coaching, or partner with celebrity academy brands (Padukone, Gopichand). The docs contain no distribution analysis: no channel, no CAC estimate, no answer to how a solo developer reaches academy buyers that VISIST.AI already serves through the Padukone network."}, {"claim": "VERIFIED ? pricing benchmarks. SwingVision \$179.99/yr (swing.vision/subscribe), Trace \$180-300/yr, Veo ~\$1,533 + \$49-149/mo all match PMF_MARKET_RESEARCH.md \$4 figures. PB Vision at \$99/yr adds a new data point at the BOTTOM of the internal \$60-180/yr B2C band ? and Game Theory bundles highlights quasi-free with court booking, capping standalone B2C highlight WTP in India."}, {"claim": "VERIFIED ? cost model. Gemini 2.5 Flash \$0.30/1M input is live pricing (ai.google.dev/gemini-api/docs/pricing), so COMMERCIALIZATION.md:122-124's ~\$0.05/10-min-clip input estimate holds; batch tier halves it; Gemini 3 Flash preview at \$0.50/1M doesn't break the model. PB Vision's \$8/hr-of-video API price vs ~\$0.30-0.60/hr Gemini COGS implies healthy gross margin room for the hosted API."}, {"claim": "TEMPERED ? SwingVision traction. Internal docs cite '200k+ monthly users' as competitor strength (PMF_MARKET_RESEARCH.md:139); the 2025 Wefunder data shows ~20k PAYING subscribers and FY2024 revenue of \$2.75M growing only ~12% YoY. The closest analog's decade-long grind to ~\$3-4M ARR is the realistic ceiling shape for this category, not a hockey stick."}, {"claim": "PARTIALLY SURVIVES ? commoditization/moat (OWNED_MODEL_IMPLEMENTATION_PLAN.md:21,46-52). Gemini 2.5 Pro already rivals fine-tuned models on coarse moment retrieval (QVHighlights), so 'upload match, get rough highlights' is wrapper-cheap for anyone; but 2025 temporal-grounding literature and F3Set confirm VLMs still miss precise boundaries on fast/frequent racket events, so the precision/eval-harness moat is real today ? with a visibly closing window. The repo's own hard rule (never trained-on-paid-LLM) is a self-imposed constraint competitors don't share."}, {"claim": "SURVIVES, NEWLY CORROBORATED ? the multi-court confound as target distribution (NEXT_STEPS.md:23-26). Market scan found no software-only solution to adjacent-court contamination; incumbents sell per-court installed cameras (PlaySight/Pixellot/Hudl Focus), which is expensive at academy scale. This is the strongest

technically-grounded differentiator in the whole thesis.", "MISSING ENTIRELY from internal docs: (a) who physically captures the video at an academy (existing CCTV vs phone-on-tripod vs new hardware) and who owns that workflow; (b) who pays ? player vs parent vs coach vs academy owner ? Indian comparables monetize via coaching/venue fees, not analysis SaaS; (c) celebrity/federation distribution as the apparent dominant GTM in Indian badminton; (d) any demand evidence for the conversational video-QA differentiator (PLATFORM_ARCHITECTURE.md §5A invests heavily in it with zero customer signal); (e) the C13 beachhead-validation interviews (COMMERCIALIZATION.md:40) remain not done ? the entire beachhead is desk research."], "opportunities": ["The confirmed gap: software-only, BYO-footage, full-match rally indexing + highlights for badminton ? especially on messy MULTI-COURT academy footage, which no vendor handles in software and incumbents solve with per-court hardware costing \$949-1,533/camera + subscriptions per court. For a 6-12 court Indian academy, a single existing camera + software is an order of magnitude cheaper."], "Partner-not-compete distribution in India: VISIST.AI (technique assessment) and Machaxi (coaching centers) lack rally-segmentation/highlight tech; a white-label or API integration (PB Vision x PodPlay is the proven template ? <https://podplay.app/blog/podplay-x-pb-vision-raising-the-ai-video-analysis-bar-for-pickleball-clubs>) converts the celebrity-distribution threat into a channel.", "API pricing surface validated externally: PB Vision charges partners \$8/hr of HD video processed; with Gemini COGS ~\$0.30-0.60/hr (or lower self-hosted), a per-hour processing API matches the repo's hosted-API architecture (PLATFORM_ARCHITECTURE.md) and avoids the consumer-app CAC war.", "Table tennis is an underrated second sport: the repo's own WASB-transfer finding (TT F1 0.909 out-of-the-box, NEXT_STEPS.md:20-23) aligns with Stupa proving an Indian TT analytics market with federation buyers; TT clubs/federations may be a less contested beachhead than badminton academies.", "BadPro+'s pay-per-match model (no subscription) is a tested monetization shape for price-sensitive India; per-video pricing fits the upload-API product better than SaaS tiers and lowers the WTP validation bar.", "Padel/pickleball vendors (Wingfield, Clutch) prove auto-highlight demand in racket sports but are all hardware-attached; a camera-agnostic engine could license INTO those ecosystems."], "risks": ["Distribution lock-up: Padukone (VISIST.AI, Machaxi) and Gopichand (Game Theory) networks are absorbing the credible paths into Indian badminton academies now, while this project has shipped no customer-facing product and done zero discovery interviews (C13, COMMERCIALIZATION.md:40 still 'Unvalidated').", "Commoditization clock: coarse highlights from long video are already a thin Gemini wrapper (\$0.30-0.50/1M video input tokens); if Gemini-class models close the precise-boundary gap before the owned model ships, the moat reduces to dataset + distribution ? and the project has little of the latter.", "SwingVision (\$10M+ raised, 20k paying subs) raised \$6M specifically to go multi-racket-sport; padel is explicit, badminton mentioned as applicable ? a direct collision in the prosumer segment within the planning horizon.", "Capture-behavior assumption untested: the entire B2C funnel assumes amateurs put a phone on a tripod; every funded competitor concluded the opposite and installed hardware or owns venues. Internal risk #1 (PMF_MARKET_RESEARCH.md:189-191) remains the thesis-killer and is cheap to test but untested.", "WTP compression in India: Game Theory delivers highlights bundled with court booking; group coaching is INR 1,500-5,000/mo; a \$600/yr (~INR 50k) academy SaaS must displace tangible coaching budget without a celebrity endorsement.", "Closest-analog growth is slow: SwingVision +12% YoY revenue at \$2.75M after ~8 years suggests this category produces niche businesses, not venture-scale ones ? fine for a solo/lifestyle business, fatal for a 'platform' ambition if that's the goal.", "Solo, pre-product vs funded field: Machaxi \$1.5M, Stupa \$3.3M, SwingVision \$10M+, Game Theory (Rainmatter) ? all shipping today; the repo is at 'scaffolding complete, no platform implementation started' (CLAUDE.md current state).", "Internal research staleness process risk: the 2026-06-07 market study missed every Indian competitor despite India being its own #1 recommendation ? competitive scanning needs to be a recurring gate, not a one-shot snapshot."], "pmf_verdict": "The thesis half-survives. What the evidence supports: real WTP exists for racket-sport AI video at \$99-180/yr consumer and per-court/per-hour B2B (SwingVision 20k paying subs; PB Vision \$99/yr + \$8/hr API; Trace/Veo club tiers); the cost model is verified and leaves wide margin; and the single most defensible claim in the repo is the one its own experiments produced ? nobody in the market handles multi-court, adjacent-game-contaminated static footage in software, while the market's only answer (per-court installed cameras) is expensive at academy scale. What the evidence breaks: the 'India badminton whitespace' framing is wrong ? Bangalore specifically is the most contested badminton-AI city on earth right now (VISIST.AI in 70+ local academies via Padukone, Machaxi funded by Padukone/Rainmatter, Game Theory by Gopichand, Stupa with BWF approval), and all of them own the distribution this project lacks. The strongest honest wedge is therefore not 'badminton AI for Indian academies' but 'rally-precise indexing of existing messy academy footage, sold per-hour as an engine/API' ? possibly THROUGH the very players that look like competitors, none of whom have this capability.\n\nFastest falsification path, in order, before any more engineering: (1) run the already-planned-but-undone C13 discovery interviews (10-15 Bengaluru academies) with two added questions ? do you already have multi-

court footage nobody watches, and would you pay per-match to have it indexed ? this tests both the wedge and the capture assumption in two weeks for \$0; (2) run a Gemini-3-Flash-wrapper baseline against the human-GT LOVO corpus (the harness already exists in backend/eval/) ? if a \$0.50/1M wrapper hits within a few F1 points of the learned prototype's 0.583, the owned-model investment is premature and the play is distribution/UX, not modeling; (3) one paid pilot with a single academy using their existing footage at a real price (even INR 500/match) beats any further desk research. The verdict: a viable niche product with a genuine, narrow technical moat and a verified cost structure ? but currently mis-aimed at a beachhead whose distribution is being captured by funded incumbents, with every demand-side assumption still resting on zero customer contact. Validate demand before the async-job slice or any owned-model milestone consumes another engineering month."}

2. user (2026-06-10T05:59:43.095Z)

```
BACKLOG.md
CODE_MAP.md
COMMERCIALIZATION.md
DEFERRED_HARDENING_PLAN.md
EVALUATION.md
GOLDEN_SET_IMPLEMENTATION.md
GOLDEN_SET_PHASE1_VIDEOS.md
GOLDEN_SET_VENDORING.md
HOW_RALLY_DETECTION_WORKS.md
NEXT_STEPS.md
OWNED_MODEL_IMPLEMENTATION_PLAN.md
PLATFORM_ARCHITECTURE.md
PMF_MARKET_RESEARCH.md
README.md
ROADMAP.md
TRACKNET_WSL_SETUP.md
VIDEO_RECORDING_GUIDELINES.md
archives
  243 docs/PMF_MARKET_RESEARCH.md
  223 docs/COMMERCIALIZATION.md
  46 docs/NEXT_STEPS.md
  165 docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md
  677 total
```

3. user (2026-06-10T05:59:51.371Z)

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1      # PMF & Market Study ? AI Highlight-Reel Tool for Self-Recorded Racket Sports
2
3      > **Research snapshot ? 2026-06-07.** Figures are cited as-of with source + year;
4      > validate live before relying on them for investment/pricing decisions. This is the
5      > market-research-of-record behind the PMF section of the living
6      > [ `COMMERCIALIZATION.md` ](COMMERCIALIZATION.md) tracker ($6). Product framing:
7      > a closed-source hosted API/SaaS (plus local mode) that ingests long-form,
8      > self-recorded, **static-camera, full-court amateur** racket-sports video and
9      > auto-detects rallies and compiles highlight reels. Primary sport: badminton;
10     > sport-generic architecture roadmapped to tennis, pickleball, table tennis, cricket.
11
12     > **Confidence convention.** Figures from named industry reports, federations, or
13     > company pricing pages are tagged with source + year. Inferred items are tagged
14     > ***(estimate)** / ***(low confidence)***. Emerging-category market sizes vary widely
15     > between vendors; ranges are shown rather than false precision.
16
17     ---
18
19     ## 1. Executive summary
20
21     The product sits at the intersection of three favorable trends: (a) a fast-growing
22     **sports analytics market** (~USD 5?6B in 2025, ~18?22% CAGR to USD 24?36B by ~2034
23     depending on vendor); (b) a structural shift to **amateur self-recording** on action
24     cameras and phones (action-camera market ~USD 6.5B in 2024, sports the largest
25     segment); and (c) **strong racket-sport participation** ? 220M+ badminton players,
```

106M tennis players (2024), and a near-vertical pickleball curve (~19.8M US players in 2024, +45.8% YoY).

The central tension is the **static-camera constraint**: the tool works best on fixed, full-court amateur footage and **fails** on broadcast/panning footage. This is both a moat (most competitors target team sports / broadcast) and a filter on the addressable user ? the customer must already set a camera on a tripod behind the court. That behavior is concentrated in **coaches/academies, clubs, and serious enthusiasts**, not the median casual player.

Recommended beachhead (strongest single): **badminton coaching academies and serious club/competitive players in India** ? a large, rapidly formalizing academy ecosystem (Khelo India tailwind), genuine fixed-angle recording behavior, existing willingness to pay for coaching, an underserved tooling gap (incumbents are team-sports/broadcast), and a direct-but-immature competitor field. The product is **native badminton-first** ? a genuine edge here. Secondary: **(2) US/global pickleball players & clubs** (capital-rich, fastest-growing, but SwingVision already strong); **(3) badminton academies/clubs in East/SE Asia + Denmark** (highest density).

Pricing clusters into hardware-bundled auto-filming (Veo, Trace, Pixellot, Spiideo) and software-only phone AI (SwingVision USD 179.99/yr; BadPro+ pay-per-use). A software-only BYO-camera model undercuts the hardware players; realistic anchors **(estimate): B2C USD 60-180/yr; B2B academy/club USD 300-1,500/yr per location.**

Top risks: (1) amateurs won't reliably set up a tripod ? TAM collapses to coaches/clubs; (2) free/near-free alternatives cap B2C WTP; (3) capture-quality variance is the core failure mode; (4) badminton's strongest markets (India, Indonesia, China) are the most price-sensitive and hardest to bill at Western SaaS prices.

2. Market sizing & trends

2.1 Sports analytics / sports-video market

- 2025 size:** ~USD 5.47B (Precedence Research) to ~USD 5.79B (Fortune Business Insights); other vendors USD 5.6-6.0B. **(2025)***
- Growth:** CAGR 15.7-22.5%. Precedence ? ~USD 29.75B by 2034 (20.63%); Fortune BI ? USD 24.03B by 2032 (22.5%); MarketsandMarkets (narrower) USD 4.75B by 2030 (15.7%); Future Market Insights ? ~USD 36.2B by 2035 (22.1%). **(reports 2024-2025)***
- Video analytics is the leading segment** within sports analytics (Grand View / Fortune BI) ? the directly relevant sub-market.
- Interpretation (estimate):** the **amateur/club auto-highlight** slice is a small but fast-growing fraction; most analytics revenue today is pro/broadcast/team-sports. The product targets the long tail the big numbers under-serve.

2.2 Action-camera / amateur recording adoption

- 2024 size:** ~USD 6.5B (GMIInsights) ? ~USD 16.9B by 2034 (~10% CAGR). **(2024-2025)***
- Sports is the largest end-use segment** of action cameras (GMIInsights, 2024).
- GoPro ~47% share, DJI ~20%** (electroIQ/industry, 2024); phones + cheap tripods are the de-facto amateur rig.
- Implication:** the capture-side behavior the product depends on is already widespread and growing ? **for the segments that record** ? validating "BYO-camera, software-only."

2.3 Participation & popularity by sport and geography

Sport	Participation signal	Source (year)
Badminton	220M+ play regularly; 160+ countries BWF est. via BadmintonBites (2023)	
Badminton	China ~80M registered, up to ~100M (largest base) industry compilations (2023)	
Badminton	India 20M+ recreational by 2023; academy boom + Khelo India industry compilations (2023)	
Badminton	Indonesia National sport; ~1M club players industry (est.)	

88 | Badminton ? Denmark | 100,000+ registered club members (Europe #1) | industry (2023) |
89 | Badminton ? Japan/Korea | Major school-club + workplace sport | industry (2023) |
90 | **Tennis** | **106M** players (2024), +25.6% over 5 yrs (87M in 2021) | ITF Global
Tennis Report (2021, 2024) |
91 | Tennis ? Asia | ~1/3 of players (33M of 87M) | ITF (2021) |
92 | **Pickleball** ? US | **19.8M** players (2024), +45.8% YoY, +311% over 3 yrs; ~6.2M
core | SFIA / Pickleheads / The Dink (2024) |
93 | Pickleball ? US courts | ~70,000 courts, +55% places to play YoY | SFIA / Pickleheads
(2024) |
94 | Pickleball ? global | IFP 78 member countries; UK +73% in 2024; ~282M played monthly
per PPA claim **(low confidence)** | various (2024?2025) |
95 | **Table tennis** | 30M+ players; 227 member associations | ITTF (current) |
96 | **Cricket** | 2.5B+ fans; 500M+ in India (mostly ***fans***, not self-recorders) | Cricket
Rise / Nielsen (2024?2025) |
97
98 - **Pickleball market value:** ~USD 2.2B (2024) ? USD 9.1B by 2034 at 15.3% CAGR
99 (Market.us/Custom Market Insights, 2024?2025).
100 - **Badminton equipment market:** ~USD 7.23B global (2023), Asia-Pacific dominant.
101
102 **Geographic read:** badminton density is overwhelmingly Asian + Denmark; tennis is
103 broad/high-income; pickleball is US-dominant, capital-rich, globalizing fast; cricket
104 is huge as ***fandom*** but its amateur self-recording + static-full-court fit is weak
105 (large field, not a fixed-frame court). For ***this product's static-camera fit***,
106 badminton/pickleball/tennis/table-tennis are the core; cricket is a long-tail roadmap
107 item, not a beachhead.
108
109 ---
110
111 **## 3. Customer segment evaluation**
112
113 | Segment | Market-size signal | Willingness to pay | Accessibility | Static-camera fit
| B2C/B2B | Verdict |
114 | ---|---|---|---|---|---|
115 | **Individual amateur/enthusiast** | Largest raw TAM (100s of millions) | **Low** ?
competes with free; ~USD 5?15/mo ceiling | High via app stores; high CAC/churn | **Mixed** ?
most won't set a tripod | B2C | Volume play, weak economics; use as funnel |
116 | **Coaches & coaching academies** | India academy boom; thousands across Asia |
High ? already sell coaching; USD 300?1,500/yr plausible | Reachable via
federations/distributors/social | **Strong** ? already film fixed-angle | B2B/B2B2C | **Best**
fit ? buyer, recorder, payer aligned |
117 | **Clubs & leagues** | Denmark 100k+; thousands of clubs Asia/EU | Medium-high; USD
500?3,000/yr | Medium ? committee sales cycle | **Strong** | B2B/B2B2C | Strong secondary;
longer cycle |
118 | **Tournament/event organizers** | Many small/regional events | Medium ? per-
event/seasonal | Medium ? episodic | **Strong** | B2B | Good per-event upsell; lumpy |
119 | **Schools & universities** | Large (esp. Asia/Japan clubs) | Low-medium ? budget-
constrained | Low ? slow procurement | Strong | B2B | Slow; grant/Khelo-India dependent |
120 | **National/regional federations** | Few, high-value | Medium-high; procurement-
heavy/political | Low ? long cycles | Strong | B2B | Lighthouse/credibility, not beachhead |
121 | **Broadcasters / streamers** | High value per deal | High | Low ? incumbents
entrenched | **Poor** ? panning = failure mode | B2B | **Avoid** ? violates core constraint |
122
123 **Key dynamics:** the static-camera constraint is a **B2B filter** ? coaches/academies/
124 clubs already record from a fixed angle, so the product fits existing behavior (cleanest
125 PMF). B2C is a large low-WTP, high-CAC funnel best used to feed B2B. Broadcasters
126 explicitly violate the constraint ? exclude.
127
128 ---
129
130 **## 4. Competitive & pricing benchmarks**
131
132 | Product | Category | Model | Price | Notes (year) |
133 | ---|---|---|---|---|
134 | **Veo (Cam 3)** | Team-sports auto-filming | Hardware + sub | Camera ~USD 1,533 (5G
~1,886) + ~£395?895/yr | 15,000+ clubs/80 countries; \$115M raised (2025) |

135 | ****Trace**** | Soccer auto-filming | ****Free camera + sub**** | USD 180/yr (Basic) ? 300 (Pro) | Camera leased free with sub (2025) |

136 | ****Pixellot (Air)**** | Auto camera/streaming | Hardware + sub | Camera ~USD 949 + USD 69?167/mo | Institutional/league; 10,000+ clubs (2025) |

137 | ****Spiideo**** | Auto-filming/analytics | Sub incl. camera | ~USD 3,800/yr (1 team) ? 6,100/yr (club) | Institutional budgets (2025) |

138 | ****Hudl**** | Team video analysis | SaaS tiers | From ~USD 400/yr | Broad team-sports incumbent (2026) |

139 | ****SwingVision**** | ****Tennis/pickleball AI, single iPhone**** | Software freemium | ****USD 179.99/yr****; free 2 hrs/mo | Closest analog; 200k+ monthly users; \$10M+ raised (2025) |

140 | ****BadPro+**** | ****Badminton AI on phone**** | ****Pay-per-use, no sub**** | Per-analysis (? "< one coaching session") | Direct badminton competitor; shot detection, heatmaps, auto-highlights (2025) |

141 | Niche pickleball/badminton apps | Various | Mixed | Varies | Fragmented; few true auto-highlight tools (2025) |

142

143 ****What buyers actually pay (synthesis):**** consumer software-only anchors at ~USD

144 100?200/yr (SwingVision 179.99); hardware-bundled USD 180?900/yr prosumer/club,

145 USD 3,800?6,100/yr institutional; badminton-specific BadPro+ uses *pay-per-use* ?

146 signaling a real badminton AI market but ****unproven subscription WTP**** (opportunity +

147 warning). ****Positioning:**** software-only + BYO-camera undercuts hardware players and

148 sells worldwide via SaaS. Realistic anchors **** (estimate): B2C USD 60?180/yr; B2B**

149 **academy/club USD 300?1,500/yr**** per location. The differentiators (cheap offline

150 self-improving detection, per-rally stats/tagging, clean licensing) support a ****B2B****

151 value story more than a B2C one.

152

153 ---

154

155 **## 5. Recommended beachhead**

156

157 Ranked by ****WTP × participation density × existing recording behavior × static-camera**

158 **fit × competitive whitespace.****

159

160 **### #1 (strongest) ? Badminton coaching academies & serious players, **India****

161 20M+ recreational players (2023); rapidly ****formalizing academy ecosystem**** + Khelo

162 India tailwind; academies already record fixed-angle; coaches are buyers who monetize

163 instruction (high relative WTP); incumbents ignore badminton; the only direct badminton

164 AI (BadPro+) is early with unproven pay-per-use. Product is *native badminton-first*.

165 ****Watch-outs:**** price sensitivity (need INR-tuned tiers); sell to the ****academy****, not

166 the individual.

167

168 **### #2 ? Serious pickleball players & clubs, **US (then UK/Australia/Asia)****

169 Fastest-growing sport (19.8M US, +45.8% YoY, 2024); capital-rich; ~70,000 courts;

170 strong recording/sharing culture; pickleball is on the roadmap. ****Watch-out:****

171 ****SwingVision already owns single-camera tennis+pickleball AI**** at USD 179.99/yr (200k+

172 monthly users) ? head-to-head differentiation, not greenfield.

173

174 **### #3 ? Badminton academies/clubs, **East/SE Asia (China, Indonesia, Malaysia, Japan,**

175 **Korea) + Denmark****

176 Highest global density (China 80M+; Denmark 100k+ structured club members; Japan/Korea

177 school-club & workplace systems), mature club infra, fixed-court recording.

178 ****Watch-outs:**** China needs local hosting/partners + is regulatory-complex; SE Asia

179 price-sensitive; Denmark/EU small but high-WTP and easy to pilot.

180

181 ****Bottom line:**** lead with ****India badminton academies**** (whitespace + behavior + buyer

182 alignment), use ****enthusiast B2C as the funnel****, validate a ****club/academy B2B2C**

183 ****badminton-Asia/Denmark**** (density).

184

185 ---

186

187 **## 6. Risks & open questions (what would invalidate the thesis)**

188

189 1. ****Tripod/static-camera adoption among amateurs.**** If serious amateurs won't set up a

190 fixed camera, B2C TAM collapses to coaches/clubs. ***Validate:*** observe 3?5 academies +

191 50?100 enthusiasts on whether they already record fixed-angle.
192 2. ****Free / near-free alternatives**** (YouTube, free app tiers, BadPro+ pay-per-use) cap
193 B2C WTP. ***Validate:*** fake-door pricing at USD 60?180/yr.
194 3. ****Capture-quality variance = the failure mode**** (gym lighting, off-center angle,
195 occlusion, mixed resolutions). ***Validate:*** benchmark rally-detection P/R on 100+ real
196 amateur clips; define a "good-capture" spec + auto-reject (ties to
197 `VIDEO\_RECORDING\_GUIDELINES.md` + the observability reports).
198 4. ****Monetization mismatch in best-participation markets**** (India/Indonesia/China are
199 price-sensitive, hard to bill). ***Validate:*** local pricing/payments pilot; per-academy
200 INR licensing.
201 5. ****Competitive response**** (SwingVision or a team-sports incumbent adds racket sports).
202 ***Validate:*** ship a badminton-quality advantage incumbents can't trivially match
203 (offline self-improving detection, per-rally tagging).
204 6. ****B2B sales-cycle drag**** for clubs/federations/schools. ***Validate:*** single-coach/
205 academy self-serve before institutional deals.
206 7. ****Clean-licensing positioning**** matters mainly to B2B buyers ? confirm it's a real
207 purchasing criterion via 5?10 buyer interviews.
208
209 ****Cheapest next steps:**** (a) 10?15 discovery interviews with Indian badminton academy
210 owners/coaches; (b) a capture-quality benchmark on real amateur footage; (c) a fake-door
211 pricing test across B2C (~USD 99/yr) and B2B academy (~USD 600/yr).
212
213 ---
214
215 ## 7. Sources
216
217 - Sports analytics market: Precedence Research; Fortune Business Insights; Grand View
218 Research; MarketsandMarkets; Future Market Insights. ***(2024?2025)***
219 - <https://www.precedenceresearch.com/sports-analytics-market>
220 - <https://www.fortunebusinessinsights.com/sports-analytics-market-102217>
221 - <https://www.grandviewresearch.com/industry-analysis/sports-analytics-market>
222 - <https://www.marketsandmarkets.com/PressReleases/sports-analytics.asp>
223 - <https://www.futuremarketinsights.com/reports/sports-analytics-market>
224 - Action camera market: [https://www.gminsights.com/industry-analysis/action-camera-](https://www.gminsights.com/industry-analysis/action-camera-market)
225 <https://www.precedenceresearch.com/action-camera-market> (2024);
226 <https://electroi.com/stats/action-camera-statistics/> (2024)
227 - Badminton participation/market: [https://badmintonbites.com/how-many-badminton-players-](https://badmintonbites.com/how-many-badminton-players-are-there-in-the-world/)
228 [https://stringsandpaddles.com/where-is-badminton-played-10-countries-where-its-](https://stringsandpaddles.com/where-is-badminton-played-10-countries-where-its-popular/)
229 <https://en.wikipedia.org/wiki/Badminton> ;
230 <https://www.marketgrowthreports.com/market-reports/badminton-market-110049> ;
231 <https://www.thesportsschool.com/top-10-badminton-academies-in-india/>
232 - Tennis: [https://www.itftennis.com/en/news-and-media/articles/itf-global-tennis-report-](https://www.itftennis.com/en/news-and-media/articles/itf-global-tennis-report-participation-hits-106-million-in-five-years/)
233 [https://sgbonline.com/itf-global-tennis-report-2021-shows-worldwide-participation-](https://sgbonline.com/itf-global-tennis-report-2021-shows-worldwide-participation-gains/)
234 [https://sfia.org/resources/as-pickleball-continues-unprecedented-growth-](https://sfia.org/resources/as-pickleball-continues-unprecedented-growth-in-every-age-group-and-region-for-third-straight-year-significant-investments-still-needed-for-court-and-facility-demand/)
235 [https://www.thedinkpickleball.com/report-pickleball-participation-surges-to-](https://www.thedinkpickleball.com/report-pickleball-participation-surges-to-nearly-20-million-in-2024/)
236 <https://www.pickleheads.com/guides/pickleball-statistics> ;
237 <https://www.custommarketinsights.com/report/pickleball-market/> ;
238 <https://market.us/report/pickleball-market/>
239 - Table tennis: https://en.wikipedia.org/wiki/International_Table_Tennis_Federation
240 - Cricket: <https://cricketrise.com/cricket-fans-in-india/> ;
241 <https://www.nielsen.com/insights/2025/global-cricket-connecting-fans-driving-roi/>
242 - Competitors & pricing: <https://www.veo.com/en-us/pricing> ; <https://traceup.com/pricing>
243 <https://www.pixellot.tv/> ; <https://www.hudl.com/pricing> ;
244 <https://swing.vision/subscribe> ;
<https://www.badproplusapp.com/>

4. user (2026-06-10T05:59:51.755Z)

```
1      # Next steps (live) ? owned-model + multisport
2
3      **Updated:** 2026-06-09 (end of the quality-iteration + multisport-recon session).
**Authoritative roadmap:**
4      `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md`. **Latest blow-by-blow:** memory `current-
state.md` / `session-handoff.md`.
5      This file = the prioritized "what to do next," refreshed each session.
6
7      ## The recommendation (one paragraph)
8      Build the **quality-first extraction harness** before anything else: a learned **TAL
head** (ActionFormer/ASFormer, MIT)
9      over trajectory features is the robust path ? it **decouples from the multi-court
"background games" confound** that the
10     unsupervised heuristic (and audio, and spatial court-gating) can't escape (all three
tested + failed; it's a *temporal*
11     rally-vs-non-rally problem). **Framework = ms-swift, base = Qwen3-VL** (Gemma-4
AMBER/bench) come **later** for the
12     conversational/grounding upgrade ? start with the head. **On-device is deferred**
(export-clean ops now so it stays
13     reachable). Everything stays **MIT/Apache-2.0/BSD** (code + data + weights). Bottleneck
= **data + per-sport trajectory
14     detectors**, not model architecture.
15
16     ## Latest findings (2026-06-09)
17     - **6-match badminton LOVO:** learned prototype **0.583** vs heuristic **0.480**
(learned leads since n=3, AUC 0.83?0.92,
18     decoupled from multi-court). The "contrast" metric (in-rally÷dead-time visibility)
predicts heuristic F1 monotonically.
19     - **?? Multisport (Roora pickleball + TT):** labels **ingest cleanly** (`curate convert-
cvat` ? 6 rallies/sport, sane).
20     **WASB-transfer SURPRISE (directional, n=6 each ? TINY):** the badminton shuttle
detector segments **TABLE TENNIS at
21     F1 0.909** out-of-the-box (!) but **pickleball only 0.308**. ? **The per-sport-
detector blocker is SPORT-DEPENDENT:** TT may
22     be servable with WASB short-term; pickleball needs a proper (MIT/Apache) ball
detector. Confirm/deny with more videos.
23     - **Owner field notes (2026-06-09):** camera height/distance **varied on purpose**
(great generalization data); **multi-court
24     adjacent-court SOUND dominates the signal** ? empirically reconfirms audio-foreground
isolation won't work + the
25     robustness-first learned-model path. Multi-court is the realistic (academy)
environment, so this *is* the target distribution.
26
27     ## Prioritized next steps
28     1. **[owner greenlight] Phase-0 (M0.x), $0/local:**
29     - **M0.4 ? Gen-0 TAL harness** (first-class): productionize `rally_seq_proto.py` ?
one-command train?eval?ship-gate** over
30     ActionFormer/ASFormer, reusing `metrics.match_intervals`. **Define the ship-gate
target** (explicit F1@tIoU; floor = beat
31     heuristic LOVO 0.480) ? "very high P/R" must be operationalized, not just "beat the
floor."
32     - **M0.1 ? corpus spine + the per-video event-index contract** (the conversational
seam: extract ? event store ? text-to-SQL).
33     - **M0.3 ? compliance:** purge Gemini-derived training labels
(`training.label_candidates`); WASB **C3** own-weights/clean-swap (gates ship).
34     2. **Multisport thread:**
35     - Ingest the Roora pickleball/TT rally CSVs into the corpus (collector `import-
local`).
36     - **Get more pickleball + TT videos** ? re-run the WASB-transfer eval to confirm/deny
TT?0.91, pickleball?0.31 (n=6 is too few).
37     - Once each sport has ?3 matches, run a **merged prototype LOVO across badminton + TT
(+ pickleball)** ? the "how would a
38     merged model do" experiment. Identify a clean MIT/Apache ball detector for sports
```

where WASB doesn't transfer (pickleball).

```

39      3. **Housekeeping:** **rename repo off "badminton" (multisport); confirm collector =
corpus/event-store system-of-record.
40
41      ## Pointers / artifacts
42      - Roadmap: `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` · Study:
`docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md`
43      (2026-06-09 banner) · Quality: `docs/archives/quality-iterations/2026-06-multivideo-
lovo/`.
44      - Eval corpus: `output/human_lovo_manifest.json` (badminton n=6). Roora CSVs:
`%TEMP%\roora\{pickleball,tabletennis}_rallies.csv`;
45      WASB-transfer trajectories: `%TEMP%\{pickleball,tabletennis}_traj.csv`. Code:
`backend/eval/{eval_partial_labels,rally_seq_proto,calibrate_local}.py`.
46      - ? No owned-model implementation until owner greenlights a milestone. Committed next
PLATFORM slice = async job model (single-tenant).
47

```

5. user (2026-06-10T06:00:09.668Z)

```

30      Priority: **P0** (do first) · **P1** · **P2** (lowest). Rows are grouped **Pending ? In-
flight ? Completed**, each sorted by priority. (C# IDs are stable identifiers ? they don't imply
order.)
31
32      | # | Workstream | Pri | Status | Where it stands | Next action / owner decision |
33      |---|---|---|---|---|---|
34      | | **? PENDING (needs action / not started)** | | | |
35      | C5 | **Output codec royalties** | **P0** | ? | Highlights are still encoded **H.264 +
AAC** (`libx264` in `backend/pipeline/stitcher/compiler.py`) ? encode-side AVC patent exposure
remains. | Switch output to **AV1 + Opus** (royalty-free) on an **Ada-class GPU (L4/L40)** for
HW encode. See §4. Decision: provision Ada-class GPU? |
36      | C9 | **Packaging / distribution** | **P0** | ? | No `pyproject.toml`; the diagnostic
`setup.py` shadows the build system. | Design in `DEFERRED_HARDENING_PLAN.md` (#7) ? **awaiting
your approval** before code. |
37      | C12 | **SaaS foundations (async/auth/tenancy)** | **P1** | ? | Sync `/api/process`
blocks on long videos; no auth/tenancy. | Multi-user/platform strategy in
`PLATFORM_ARCHITECTURE.md` (pluggable storage/compute, tenancy, orchestration); async slice =
`DEFERRED_HARDENING_PLAN.md` (#16). |
38      | | **? IN-FLIGHT (in progress / watch)** | | | |
39      | C4 | **Runtime AI strategy (A/B/C + owned model)** | **P0** | ? | Live = **(C) Gemini
2.5 Flash, paid tier**. Long-term = **own a self-hosted generative video-QA model** (fine-tuned
open VLM) as primary, **paid Gemini always-ready as fallback** via the provider registry. | Make
"provider registry + paid fallback" a core principle; build the owned model **after** the
scaffolding/flywheel are solid. See `PLATFORM_ARCHITECTURE.md` §5A. |
40      | C13 | **Beachhead validation** | **P0** | ? | Market research
(`PMF_MARKET_RESEARCH.md`) recommends **India badminton academies** as the #1 beachhead.
Unvalidated. | Cheapest tests: 10?15 academy/coach discovery interviews; rally-detection P/R
benchmark on 100+ real amateur clips; fake-door pricing (B2C ~$99/yr, B2B academy ~$600/yr). See
§6. |
41      | C8 | **Golden-set vendoring spend** | **P1** | ? | ADR-006: **Roora (Bengaluru)**
selected for an initial paid pilot; licensed/owned video only (no user uploads to vendors). |
Confirm quality-per-dollar from the pilot; then GS-3 `HumanVendorProvider`. See
`GOLDEN_SET_VENDORING.md`. |
42      | C3 | **Shuttle-tracking weights provenance** | **P1** | ? ? | WASB-SBDT **code is MIT
(clear)**; distributed **checkpoints are academic-data-trained** ? not covered by the MIT code
grant. MonoTrack/YOLO-NAS banned (noncommercial). | Train-own weights (Phase-4 roadmap) **or**
confirm per-checkpoint terms ?? before commercial deploy. |
43      | C14 | **Owned-model base licensing + training-data policy** | **P1** | ? ? | Owned
video-QA model (PLATFORM §5A) must build on **Apache-2.0-clean, video-capable** bases ?
**Qwen2.5-VL-7B/32B** (Apache; 3B/72B are *not*), **Qwen3-VL-4B/8B**, or **Gemma 4** ? to keep
weights private, gate, and monetize. **Reject Gemma 3** (viral derivative clause ? permanent
Google PUP), Llama Vision (naming + 700M-MAU), `gpt-oss` (text-only). **Hard rule: never train
on paid-LLM (Gemini/OpenAI/Anthropic) outputs** (competing-model clause). The moat is the
**consented dataset + eval harness**, not the model. | Lock to Apache-2.0; **open-model teachers
only**; separate explicit **training opt-in**; **exclude minors** from the training pool; per-
clip consent ledger. See PLATFORM §5A/§7. |

```

```

44      | C6 | **Inbound HEVC decode** | **P2** | ? ?? | Decoding users' GoPro H.264/**HEVC**
uploads is unavoidable; HEVC pools are fragmented + highest-uncertainty. | Low/uncertain
residual; confirm decode-side exposure with counsel before launch. |
45      | C7 | **Hosting unit-economics** | **P2** | ? | Per-video cost is *cheap* for Gemini
(~$0.05 input for a 10-min clip); risk is volume × latency, not unit price. Self-host/offline
costs unmeasured. | **Benchmark $/video & latency on the target VM** for A/B/C before pricing.
See §5. |
46      | C10 | **Value layer (product surface)** | **P2** | ? | Minimal exports shipped
(EDL/JSON/CSV + `basic_stats`, PR #51). | Deepen: per-rally tags/notes that feed golden, per-
player/match stats, clip gallery. See §6. |
47      | C11 | **Input-quality enforcement** | **P2** | ? | Per-video observability reports
surface a customer verdict + footgun flags incl. audio-present (PR #50). | Turn more
`VIDEO_RECORDING_GUIDELINES` dimensions into validators/report rules. |
48      | C2 | **`supervision` ByteTrack pin** | **P2** | ? | Pinned `<0.30.0` (deprecation
ahead). | Migrate before lifting the pin (BACKLOG). Not a legal risk, a maintenance one. |
49      | | **? COMPLETED** | | | |
50      | C1 | **AGPL / copyleft removal** | ? | ? | ADR-005 shipped: AGPL `ultralytics` YOLOv8
replaced with **torchvision Faster R-CNN (BSD) + `supervision` ByteTrack (MIT)**. No AGPL
code/weights in the stack. | Done. Keep new deps permissive. |
51
52      ---
53
54      ## 2. Licensing & dependency posture (refreshed risk table)
55
56      Verdicts updated against the current stack (`requirements.txt` + code). Original
57      2026-05-30 verdicts and the 25-claim verification transcript are in
58      `archives/MONETIZATION_AUDIT.md`.
59
60      | Dependency | License | SaaS obligation? | Current verdict | Notes |
61      |---|---|---|---|---|
62      | ~`ultralytics` + `yolov8n.pt`~ | AGPL-3.0 | Yes (§13) | ? **REMOVED** | Replaced per
ADR-005 ? no longer in the stack. |
63      | **torchvision** detectors (Faster R-CNN/RetinaNet) | BSD-3 | No | ? LIVE | The chosen
replacement; person = COCO class (see `yolo_hybrid.py` `_PERSON_CLASS_ID`). |
64      | **`supervision`** (ByteTrack) | MIT | No | ? LIVE | Pinned `<0.30.0` ? migrate before
bump (C2). |
65      | **WASB-SBDT** (shuttle) | MIT (code) | No | ? code clear | Weights provenance open
(C3). |
66      | **MonoTrack / YOLO-NAS** | noncommercial | ? | ? BANNED | Do not use. |
67      | **Gemini API** (paid tier) | service terms | Paid: no training/review | ? LIVE WATCH |
Must bill the paid tier to keep user video private (C4). |
68      | H.264 / HEVC / AAC codecs | patent pools | royalties on encode/decode | ? ?? |
Separate from ffmpeg's license. Encode-side open (C5); decode-side residual (C6). |
69      | ffmpeg/ffprobe, google-genai, torch, opencv, numpy,
fastapi/uvicorn/pydantic/jinja2/dotenv | LGPL/Apache/BSD/MIT | No | ? CLEAR | Invoked as
subprocess / imported; all permissive. |
70      | **`obsreport`** (reports) | permissive (private) | No | ? soft dep | Pin the
`git+ssh@vX` tag once the repo is published (BACKLOG). |
71
72      **Bottom line:** the two hard blockers from the audit (Ultralytics; banned weights)
73      are resolved/avoided. Remaining commercial-legal items are **codec royalties (C5/C6)**
74      and **weights provenance (C3)**, plus the metered-API privacy posture (C4).
75
76      ---
77
78      ## 3. Runtime AI strategy ? A/B/C
79
80      The semantic rally-boundary step is the one external/metered dependency. Three paths
81      (from the audit), with current standing:
82
83      - ** (A) Eliminate ? offline learned segmenter.** ActionFormer (MIT) / our own learned
84      classifier on `candidate_segments.stats`. **Best steady-state margin** (no per-call
85      cost, no external dep) and matches the Phase-4 roadmap (substrate = WASB
86      shuttle-motion, decided in ADR-004). **Most engineering.** ? **target**
87      - ** (B) Self-host VLM.** Qwen2.5-VL-7B/32B (Apache-2.0) or InternVL2.5-8B (MIT). No

```

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88     per-call fees; pays in GPU VRAM (~16?24 GB for 7B). ?? Qwen 3B is noncommercial.
89 - *(C) Keep Gemini 2.5 Flash (paid).* Fastest, cheap per video, video-native;
90     ongoing metered cost + external dependency. **? current live path.**
91
92 Hosted alternatives for (C) comparison: Anthropic Claude (no training on
93 inputs/outputs; customer owns outputs; **no native video** ? weaker here);
94 OpenAI (no training by default; ZDR available); Gemini remains strongest video-native.
95
96 ---
97
98 ## 4. Codec / output economics (the live gap ? C5)
99
100 **The trap:** ffmpeg being free (LGPL) does **not** grant the *patent* rights to the
101 codecs it implements. Royalties attach to the act of encode/decode.
102
103 - **Encode side (we control it):** highlights currently emit **H.264/AAC**. AVC (Via LA)
104     is first 100k units/yr royalty-free with a permanent "free to end-users over the
105     internet" provision ? a small SaaS likely falls under thresholds, but ?? confirm.
106 - **Mitigation:** emit **AV1 (SVT-AV1, BSD + AOMedia royalty-free patent grant) + Opus
107     (royalty-free)**. NVENC AV1 HW-encode needs **Ada-class GPUs (L4/L40)**; Ampere/Turing
108     must use **SVT-AV1 (CPU)**.
109 - **Decode side (unavoidable, C6):** inbound GoPro **HEVC** decode ? fragmented pools,
110     highest uncertainty; low residual.
111
112 **Recommendation:** provision an **Ada-class GPU (L4/L40)** and standardize highlight
113 output on **AV1 + Opus** ? removes the encode-side royalty question almost entirely.
114 *Implementation lives in `backend/pipeline/stitcher/compiler.py` (codec args).*
115
116 ---
117
118 ## 5. Hosting unit-economics (C7)
119
120 Reference workload: ~30-min 1080p match on a cloud NVIDIA GPU VM.
121
122 - *(C) Gemini 2.5 Flash (paid):** input $0.30/1M tokens, output $2.50/1M; video ~263
123     tok/s ? 10-min clip ? 158k input tokens ? **~$0.05 input**. Cheap per unit; the
124     economic risk is **volume × latency**. Flash-Lite is cheaper; Pro ~10×.
125 - *(B) Self-host Qwen2.5-VL-7B:** $0 API; cost = GPU-hours / throughput; needs ?24 GB
126     VRAM (L4/A10). $/video vs Gemini is **open ? measure**.
127 - *(A) Offline ActionFormer/learned:** cheapest at inference; cost = one-time training
128     + light GPU. **Best steady-state margin.**
129
130 > The A/B/C $/video comparison depends on GPU instance + batch throughput ?
131 > **benchmark on the target VM before pricing the product.**
132
133 ---
134
135 ## 6. Product-market fit & the quality/data phase
136
137 ### Beachhead (from market research, 2026-06-07 ? see
138 [`PMF_MARKET_RESEARCH.md`](PMF_MARKET_RESEARCH.md))
139 The static-camera constraint is a **B2B filter**: the buyer must already record from a
140 fixed angle, which points away from the median casual player and toward coaches/clubs.
141 Ranked beachheads:
142 1. **Badminton coaching academies & serious players, India** *(strongest)* ? 20M+
143     players,
144     formalizing academy ecosystem (Khelo India), genuine fixed-angle recording, coaches
145     who already monetize, badminton whitespace vs incumbents. Sell to the **academy**,
146     not
147     the individual; needs INR-tuned pricing.
148 2. **US/global pickleball players & clubs** ? fastest-growing sport (19.8M US, +45.8%
149     YoY)
150     but **SwingVision already owns single-camera tennis/pickleball AI** (USD 179.99/yr) ?
151     head-to-head, not greenfield.
152 3. **Badminton academies/clubs in East/SE Asia + Denmark** ? highest density; China

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needs
149     local hosting/partners; Denmark/EU small but high-WTP, easy to pilot.
150
151     Pricing anchors *(estimate)*: software-only BYO-camera **B2C USD 60?180/yr**, **B2B
152     academy/club USD 300?1,500/yr** per location (undercuts hardware-bundled Veo/Pixellot/
153     Spiideo; comparable to SwingVision software-only).
154
155     ### PMF risks (from the readiness review)
156     - **Capture quality is the gate.** Accuracy is capture-conditional; broadcast/handheld
157     footage is a failure mode. The wedge (static, tripod, full-court) is defensible but
158     narrows the addressable market ? enforce/guide it (C11, `VIDEO_RECORDING_GUIDELINES`).
159     - **Value beyond raw segments is table stakes.** Time-saved editing alone is weak;

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6. user (2026-06-10T06:00:10.305Z)

```

1     # Owned-model implementation plan (roadmap) ? DRAFT for owner review
2
3     **Status:** PLAN / roadmap. No owned-model *implementation* starts until the owner
greenlights a specific
4     milestone. Paid Gemini stays the always-ready inference/fallback throughout (never a
training source).
5     **Grounded in:** `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md`, the n=6 LOVO +
learned prototype
6     (`docs/archives/quality-iterations/2026-06-multivideo-lovo/`), and a verified strategy
workflow (2026-06-09).
7
8     ## Design principle (owner directive): *envision now, build later*
9     Every deferred piece (conversational-QA, on-device, multisport, the VLM) must be
**architecturally envisioned
10    now** so we never hit a mammoth refactor. We **design the seams** (data contracts,
export boundaries, the
11    event store) up front and **implement behind them** when each milestone is greenlit.
Punting a *feature* is fine;
12    punting its *seam* is not.
13
14    ## North Star (reframed per PR #61 review)
15    - **Immediate deliverable:** a **training framework + harness** that produces a model
with **very high precision/
16    recall for extracting highlights + rallies from a clip**. *Quality is the first-order
objective.*
17    - **Eventual product:** a **single, sport-agnostic, conversational** video model ?
upload a clip, then ask
18    contextual questions ("longest rally", "make a clip of all service faults", "rallies
over 20 shots").
19    - **On-device is LATER** (a model-compression goal so power users save upload
cost/hassle) ? definitely on the
20    roadmap, but **not** the immediate target. Don't over-index the strategy on it now.
21    - **The moat = the consented dataset + the eval harness, not the weights.**
22
23    ## Hard guardrails (non-negotiable ? "never corrupted")
24    - **Commercial-clean licensing for EVERY dependency ? code, data, AND model weights: MIT
/ Apache-2.0 / BSD only.**
25    (Ratifies the prior "Apache-2.0 only" to include MIT/BSD.) Reject viral/NC/custom-
restrictive: Gemma 3 (viral),
26    **Gemma 3n** (Gemma Terms follow derivatives ? even cleaner reject), Llama 4 (700M-MAU
cap), Commons-Clause repos.
27    - **Never train on paid-LLM (Gemini/OpenAI/Anthropic) outputs.** Also forbidden: public
grounding-CoT datasets that
28    were Gemini-generated (e.g. TVG-R1's 56K) ? we must originate our own CoT supervision
(a real, binding cost).
29    - **Exclude minors; per-user training data needs a separate explicit opt-in; split data
by match.**
30    - ? **Base-model pretraining provenance is unauditabile for *every* open VLM (incl.
Qwen3-VL).** Our "no paid-LLM
31    training" rule is enforceable at *our fine-tuning data* layer, not the base's

```

```

pretraining ? an **explicit owner
32     risk-acceptance** is required to use any open VLM. (Governed by the base's license,
not our pipeline.)
33
34     ## The model strategy (verified 2026-06-09)
35     **Framework ? base** (the Gemma-4-vs-ms-swift clarification): a *framework* is the forge
(run SFT/GRPO, emits
36     weights); a *base* is the metal you forge. You pick one framework and feed it any clean
base.
37     - **Framework: ms-swift (Apache-2.0)** ? the one forge that does native **video + audio
+ timestamp-grounding +
38     GRPO/RFT with a custom video reward** in one stack. Keep **unsloth** (Apache-2.0 core;
*avoid its AGPL Studio UI*)
39     as the 8 GB-local SFT/compression tool for the deferred on-device goal. ? Point-in-
time snapshot ? re-verify at impl.
40     - **Base (when we reach the VLM): Qwen3-VL (Apache-2.0, uniform across sizes)** ? long-
video context, purpose-built
41     **timestamp emission** (second-level localization), multi-turn video chat. Re-confirm
the exact size's LICENSE
42     pre-release. **Gemma-4 stays AMBER and bench-only** ? its Apache grant is likely but
its Prohibited-Use posture is
43     unconfirmed (counsel needed), *and* its video/timestamp capability is disputed; not
the primary extractor.
44     **InternVL3 (MIT/Apache)** = clean fallback if a Qwen blocker appears. ? base swap is
cheap *only within
45     timestamp-capable Apache bases* (timestamp data-format + grounding-CoT are base-
specific).
46     - **START WITH THE TAL HEAD, NOT THE VLM** (the decisive call): fine-tuned VLMs **miss
strict temporal boundaries**
47     (best RL-post-trained video VLM ? R@0.7 50% on open-domain benchmarks; "coarse
regions, not accurate boundaries").
48     **Boundary accuracy IS the product.** The head (`rally_seq_proto.py`) is already de-
risked on our corpus (held-out
49     AUC 0.83?0.92), trains in minutes at ~4.5 GB on the 2070 Super ($0), is ~1M params
(trivially compressible later),
50     and **doubles as a pseudo-label teacher + the honest baseline the VLM must beat.** So:
**head now ? VLM later**,
51     strictly sequenced. When the VLM comes, prefer **two-stage** (VLM coarse-proposes,
head refines boundaries) over
52     wholesale replacement. *(Open: the VLM-ceiling number is open-domain; our amateur
footage may differ ? re-measure.)*
53
54     ## Conversational video-QA ? build the bookkeeping NOW, punt the model
55     **Punt the conversational *feature*** until extraction quality is solid (current LOVO
~0.583 ? "very high"; a chat UI
56     over a weak extractor exposes the weakness). **Reject end-to-end-VLM-answers-each-turn**
(expensive; bad at exact
57     counting; the strong teachers are license-forbidden). **But build the *seam* now** (the
brief requires it):
58     - **Architecture = "extract once, query many":** EXTRACTION (TAL head ? later VLM) emits
structured timestamped
59     spans ? a durable **per-video EVENT STORE** ? a **structured QUERY layer** (text-to-
SQL + ffmpeg clip-cut). The
60     cited queries ("longest rally", "all service faults", "rallies > 20 shots") are
**structured-numeric ? SQL, not a
61     VLM job** (SQL counts exactly where VLMs hallucinate).
62     - **Designed-now seam:** make the **per-video event-index schema a first-class data
contract** alongside the M0.1
63     corpus spine: per rally `{video_id(MD5), rally_ordinal, sport, start, end, shot_count,
ending_reason/fault_tags,
64     derived_clip_ref, provenance, confidence}`. **Reserve the event/fault-tag taxonomy
(e.g. `service_fault`) up front**
65     even if unpopulated. ? This is a real DB delta (a `PRAGMA user_version` migration):
`play_segments` today has
66     `video_id/start_time/end_time/duration/server/receiver/metadata` only; `highlights`

```

isn't a table yet ? so it's an
67 *extension with new columns + a highlights table*, not a no-op. **Punt** the NL
parser/agent until quality clears the bar.
68 - Paid Gemini may answer genuinely open-ended edge questions **at inference** over our
structured index (allowed ?
69 it reasons over our data, it is not trained on its outputs).
70

7. assistant (2026-06-10T06:00:23.571Z)

Internal line citations verified. Now web-checking the load-bearing external claims.

8. user (2026-06-10T06:00:29.666Z)

Web search results for query: "Machaxi funding Rainmatter Prakash Padukone badminton AI coaching \$1.5 million"

Links: [{"title": "Machaxi Secures \$1.5 Million to Expand AI-Powered Badminton Coaching Across India", "url": "https://india.entrepreneur.com/news-and-trends/machaxi-secures-15-million-to-expand-ai-powered-badminton/493070"}, {"title": "Machaxi Raises \$1.5 Million to Scale AI-Powered Badminton Coaching Across India ? Outlook Business", "url": "https://www.outlookbusiness.com/start-up/investors/machaxi-raises-15-million-to-scale-ai-powered-badminton-coaching-across-india"}, {"title": "Machaxi Raises \$1.5M to Revolutionize Grassroots Badminton with AI and Prakash Padukone", "url": "https://www.tice.news/tice-dispatch/filling-the-sports-gap-rainmatter-funds-ai-sports-startup-machaxi-9353716"}, {"title": "Machaxi Secures \$1.5 Million Investment to Expand AI-Powered Badminton Coaching & Grassroots Sports Development", "url": "https://startupsuccessstories.in/machaxi-secures-1-5-million-investment-to-expand-ai-powered-badminton-coaching-grassroots-sports-development/"}, {"title": "Machaxi Raises \$1.5M from Prakash Padukone, Rainmatter and Others to Expand AI-Based Grassroots Badminton Training - Passionate In Marketing Machaxi Raises \$1.5M to Expand AI-Powered Grassroots Badminton Coaching Across India", "url": "https://www.passionateinmarketing.com/machaxi-raises-1-5m-from-prakash-padukone-rainmatter-and-others-to-expand-ai-based-grassroots-badminton-training/"}, {"title": "Machaxi aces \$1.5m fund raise from Padukone and Nithin Kamath's Rainmatter | 1 Indian Television Dot Com", "url": "https://www.indiantelevision.com/mam/digital/machaxi-aces-1.5m-fund--raise-from-padukone-and-nithin-kamath%E2%80%99s-rainmatter-250611"}, {"title": "Machaxi raises \$1.5m to expand AI-based grassroots badminton | Sportzpower", "url": "https://sportzpower.com/machaxi-raises-1-5m-to-expand-ai-based-grassroots-badminton/"}, {"title": "Machaxi-Padukone School of Badminton to set up centre in Hyderabad - Telangana Today", "url": "https://telanganatoday.com/machaxi-padukone-school-of-badminton-to-set-up-centre-in-hyderabad"}, {"title": "Machaxi raises \$1.5mn from Prakash Padukone, Rainmatter and others - MediaBrief", "url": "https://mediabrief.com/machaxi-raises-1-5m/"}]

Based on the web search results, here's the information about Machaxi's funding:

Machaxi Secures \$1.5 Million Funding

Machaxi, a Bengaluru-based sports-tech startup focused on grassroots coaching, has raised \$1.5 million in funding to scale its AI-powered training model and expand operations into three more cities?Hyderabad, Pune, and Chennai.

Funding Round Details

The funding round was led by Rainmatter, the investment arm of Zerodha, and included participation from Indian badminton legend Prakash Padukone as well as existing backers.

Strategic Initiative

Machaxi is collaborating with the Padukone School of Badminton to launch the "Machaxi x Padukone School of Badminton" initiative, which aims to set up over 1,000 coaching centers across India within the next four years.

AI-Powered Coaching System

The company's AI coaching system is designed not to replace human coaches, but to standardize training quality, assist in performance tracking, and ensure that the coaching model remains scalable.

The funding was announced in June 2025, reflecting strong investor confidence in Machaxi's mission to democratize access to quality badminton coaching across India through technology-driven solutions.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. user (2026-06-10T06:00:31.305Z)

Web search results for query: "VISIST.AI badminton Padukone School of Badminton skill assessment Bengaluru academies"

Links: [{"title": "VISIST.AI partners with Padukone School of Badminton to digitise athlete skill assessments", "url": "https://theprint.in/india/visist-ai-partners-with-padukone-school-of-badminton-to-digitise-athlete-skill-assessments/2716417/"}, {"title": "VISIST.AI Partners with Padukone School of Badminton to Revolutionize Athlete Training | Sports-Games", "url": "https://www.devdiscourse.com/article/sports-games/3533385-visistai-partners-with-padukone-school-of-badminton-to-revolutionize-athlete-training"}, {"title": "Indian badminton players use artificial intelligence to up their game - Mugglehead Investment Magazine", "url": "https://mugglehead.com/indian-badminton-players-use-artificial-intelligence-to-up-their-game/"}, {"title": "Ananya Verma - AI Enthusiast | Exploring the world of Intelligent Technology | LinkedIn", "url": "https://www.linkedin.com/in/ananya-verma-6339b1251/"}, {"title": "Padukone School of Badminton", "url": "https://karthiks-testing-site.webflow.io/"}, {"title": "Bangalore Badminton Courts ? Padukone School Of Badminton", "url": "https://padukoneschoolofbadminton.com/pages/bangalore-badminton-courts"}, {"title": "About Padukone School of Badminton - Vision & Mission ? Padukone School Of Badminton", "url": "https://padukoneschoolofbadminton.com/pages/about-padukone-school-of-badminton"}, {"title": "Padukone School Of Badminton", "url": "https://padukoneschoolofbadminton.com/"}, {"title": "Padukone School of Badminton Expands to 75 Centres in First Year, Targets 250 by 2027 | Entrepreneur", "url": "https://www.entrepreneur.com/en-in/news-and-trends/padukone-school-of-badminton-expands-to-75-centres-in-first/493030"}, {"title": "PPSM", "url": "http://www.ppsm.org.in/children-coaching-programs.html"}]

Based on the search results, here's what I found about VISIST.AI, badminton, Padukone School of Badminton, and skill assessment in Bengaluru:

Partnership Overview

VISIST.AI, a sports tech start-up incubated by the FSID at the Indian Institute of Science, announced a tie-up with the Padukone School of Badminton to digitise skill assessments across the academy's network.

Technology and Capabilities

VISIST.AI leverages artificial intelligence to deliver performance insights, skill assessments, and data-driven coaching for athletes and sports academies. More specifically, VISIST.AI's program will inform the player about flaws in their posture, grip, swing velocity and overall technique.

Through this partnership, VISIST.AI's Visual Intelligence platform will be deployed across PSB's academy ecosystem to enable data-driven athlete evaluations, personalised feedback, and evidence-based training insights.

Scale in Bengaluru

More than 70 academies in the city have been employing the technology, many of which are overseen by the Padukone School of Badminton.

Key Leadership Perspective

VISIST.AI aims to deliver affordable and scalable solutions that bring elite-level performance intelligence to grassroots and para-athletes using everyday devices like smartphones.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

10. user (2026-06-10T06:00:44.480Z)

Web search results for query: "SwingVision Wefunder revenue 2024 paying subscribers ARR kingscrowd"

Links: [{"title": "SwingVision on Wefunder 2025 - Kingscrowd", "url": "https://kingscrowd.com/swingvision-on-wefunder-2025/"}, {"title": "SwingVision on Wefunder 2024 - Kingscrowd", "url": "https://kingscrowd.com/swingvision-on-wefunder-2024/"}, {"title": "SwingVision Stock Price, Funding, Valuation, Revenue & Financial Statements", "url": "https://www.cbinsights.com/company/swingvision/financials"}, {"title": "Annual Recurring Revenue (ARR) - Wefunder", "url": "https://help.wefunder.com/basic-jargon/annual-recurring-revenue-(arr)"}, {"title": "SwingVision: Automated Stats, Highlights & Officiating Using Just Your Phone | Wefunder, Home of the Community Round", "url": "https://wefunder.com/swingvisioncf2025"}, {"title": "Wefunder Updates, Team, and Funding Progress | Wefunder, Home of the Community Round", "url": "https://wefunder.com/wefunder"}, {"title": "Swing Vision - 2025 Funding Rounds & List of Investors - Tracxn", "url": "https://tracxn.com/d/companies/swing-vision/_mtjZ2V0bNZXqkMo6dQLL9Cod9rbXNENVpAULBpYUGHk/funding-and-investors"}, {"title": "SwingVision - Crunchbase Company Profile & Funding", "url": "https://www.crunchbase.com/organization/swingvision"}, {"title": "Series A - SwingVision - 2023-10-18 - Crunchbase Funding Round Profile", "url": "https://www.crunchbase.com/funding_round/swingvision-series-a--3eba2a17"}]

Based on the search results, here's what I found about SwingVision's financial metrics from their Wefunder fundraising campaign:

2024 Financial Performance

In fiscal year 2024, SwingVision recorded \$2,746,593 in revenue, representing a 12% year-over-year increase from \$2,455,089 in 2023.

Paying Subscribers and ARR

By the end of 2024, SwingVision reported more than 20,000 paying subscribers. The company has over \$4 million in annual recurring revenue (ARR), with a base of 20,000 subscribers contributing to this figure, which suggests healthy average revenue per user.

Growth Context

FY2023 revenue had roughly doubled from just over \$1 million in 2022, indicating the company scaled rapidly after launch and is now entering a steadier growth phase.

This information comes from Kingscrowd's analysis of SwingVision's Wefunder crowdfunding campaign, which provides detailed breakdowns of the company's financial metrics and business performance.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

11. user (2026-06-10T06:00:45.647Z)

Web search results for query: ""PB Vision" pickleball API pricing "\$8" per hour video partner"

Links: [{"title": "PB Vision API Partner Guide - PB Vision", "url": "https://help.pb.vision/articles/2793895-pb-vision-api-partner-guide"}, {"title": "PB Vision - Pickleball analytics powered by AI", "url": "https://pb.vision/"}, {"title": "API License Agreement for Pickleball Vision AI Inc.", "url": "https://pb.vision/api-license-agreement"}, {"title": "PB Vision: Pricing, Free Demo & Features | Software Finder", "url": "https://softwarefinder.com/artificial-intelligence/pb-vision"}, {"title": "Court Insight by PB Vision - PB Vision", "url": "https://help.pb.vision/articles/9341690-court-insight-by-pb-vision"}, {"title": "Get Your DUPR Rating with AI & PB Vision - thePicklebase | ThePickleBase", "url": "https://www.thepicklebase.com/coaching-tips/future-pickleball-ratings-getting-dupr-rating-power-ai"}, {"title": "PB Vision - App Store - Apple", "url": "https://apps.apple.com/us/app/pb-vision/id6467020610"}, {"title": "DUPR Blog | DUPR and PB Vision Partner to Shape the Future of AI Pickleball Ratings", "url": "https://www.dupr.com/post/dupr-and-pickleball-vision-partner-to-shape-the-future-of-ai-pickleball-ratings"}]

Based on the search results, here's what I found about PB Vision's API pricing:

PB Vision API pricing is \$8 per hour of video processed for HD (1080p or lower) content, which equals about \$1.60 per 12-minute game. For 4K video, the cost is \$12 per hour (about \$2.40 per 12-minute game).

The PB Vision API allows approved partners to send game footage and receive AI-generated performance data back once processing is complete. The AI detects rallies, shots, errors, player movement, and player ratings, delivering pro-level analytics that users can explore in PB Vision or inside their own experience.

API minutes cost more than subscription minutes because API uploads are transcoded in the cloud at PB Vision's expense, whereas regular subscription users get video transcoding on their phone at lower cost, and some subscription users don't use all their minutes, allowing lower average pricing.

API partners receive additional benefits including API uploads and webhooks, priority support, whitelabeling options, and premium insight reports for non-subscribers.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

12. user (2026-06-10T06:01:00.906Z)

Web search results for query: "Gemini API pricing 2.5 Flash \$0.30 per 1M input tokens video Gemini 3 Flash pricing"

Links: [{"title": "Gemini API Pricing May 2026: 3.5 Flash, 3.1 Pro, 2.5 Lite", "url": "https://www.metactoo.com/blogs/the-true-cost-of-google-gemini-a-guide-to-api-pricing-and-integration"}, {"title": "Gemini 3.5 Flash - API Pricing & Benchmarks | OpenRouter", "url": "https://openrouter.ai/google/gemini-3.5-flash"}, {"title": "Google Gemini API Pricing 2026: Gemini 3.5 Flash, 3.1 Pro & 2.5 Models - DevTk.AI", "url": "https://devtk.ai/en/blog/gemini-api-pricing-guide-2026/"}, {"title": "Gemini API Pricing", "url": "https://developer.puter.com/tutorials/gemini-api-pricing/"}, {"title": "Agent Platform Pricing | Google Cloud", "url": "https://cloud.google.com/gemini-enterprise-agent-platform/generative-ai/pricing"}, {"title": "Gemini API Pricing Calculator & Cost Guide (Jun 2026)", "url": "https://costgoat.com/pricing/gemini-api"}, {"title": "Google's Gemini 3.5 Flash costs 3x the model it replaced, and the era of cheap AI is ending", "url": "https://www.xda-developers.com/google-gemini-3-5-flash-costs-3x-model-replaced-cheap-ai-ending/"}, {"title": "Gemini 3.5 Flash API Pricing (May 2026): \$1.50/\$9.00 per 1M Tokens", "url": "https://devtk.ai/en/models/gemini-3-5-flash/"}, {"title": "Gemini 3.5 Flash Pricing: Token Costs & Budget Examples (2026)", "url": "https://evolink.ai/blog/gemini-3-5-flash-pricing-guide"}, {"title": "Gemini Developer API pricing | Gemini API | Google AI for Developers", "url": "https://ai.google.dev/gemini-api/docs/pricing"}]

Based on the search results, here's the information about Gemini API pricing:

Gemini 2.5 Flash Pricing

Google currently lists standard Gemini 2.5 Flash text/image/video pricing at \$0.30/M input and \$2.50/M output. This matches the \$0.30 per 1M input tokens mentioned in your query.

Gemini 3 Flash Pricing

Gemini 3 Flash costs \$0.50 and \$3 per million tokens for input and output respectively.

Gemini 3.5 Flash Pricing

Gemini 3.5 Flash is listed as a stable model with standard pricing at \$1.50/M input and \$9.00/M output, and cached input at \$0.15/M.

Video Pricing

Regarding video specifically, Gemini Embedding 2 costs \$0.20 per 1M text tokens, \$0.45 per 1M image tokens, \$6.50 per 1M audio tokens, and \$12.00 per 1M video tokens. However, for the main Gemini text/image/video models, the pricing mentioned above applies uniformly across all input types.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

13. user (2026-06-10T06:01:07.141Z)

Web search results for query: "Game Theory sports startup Bengaluru smart courts highlights Gopichand Rainmatter; Stupa Sports Analytics badminton BWF funding"

Links: [{"title": "GameTheory - 2026 Company Profile, Team, Funding, Competitors & Financials - Tracxn", "url": "https://tracxn.com/d/companies/gametheory/___pd0B46xo2J1QH-rpu1vpiw-gl-kIJXu3VNWhc9otULM"}, {"title": "Using computer vision, Game Theory helps sports players ...", "url": "https://yourstory.com/2024/02/computer-vision-game-theory-sports-tech-startup-nithin-kamath-rainmatter"}, {"title": "Sportstech Startup Game Theory Bags Funding From Nithin Kamath, Rohan Bopanna", "url": "https://inc42.com/buzz/game-theory-bags-funding-from-nithin-kamath-rohan-bopanna-to-provide-sports-technology/"}, {"title": "Sports Tech Startup Game Theory Secures Funding from Badminton Legend Pullela Gopichand and Others | Startup Story", "url": "https://startupstorymedia.com/insights-sports-tech-startup-game-theory-secures-funding-from-badminton-legend-pullela-gopichand-and-others/"}, {"title": "Game Theory Ropes In Chief Badminton Coach Pullela Gopichand, Others As Investors", "url": "https://inc42.com/buzz/sportstech-startup-game-theory-ropes-in-chief-badminton-coach-pullela-gopichand-others-as-investors/"}, {"title": "Bengaluru-Based Game Theory Raises Funding From Several Prominent Sports Personalities - Vetted Sports", "url": "https://vettedsports.com/bengaluru-based-game-theory-raises-funding-from-several-prominent-sports-personalities/"}, {"title": "Bengaluru-based Game Theory raises funding from prominent sports personalities", "url": "https://indianstartupnews.com/funding/bengaluru-based-game-theory-raises-funding-from-prominent-sports-personalities-6710948"}]

Now let me search for information about Stupa Sports Analytics:

Links: [{"title": "Stupa Sports Analytics Secures Rs 280 Million in Pre-Series A Funding Round ? Outlook Business", "url": "https://www.outlookbusiness.com/corporate/stupa-sports-analytics-secures-rs-280-million-in-pre-series-a-funding-round"}, {"title": "Stupa Sports Analytics Nets INR 28 Cr To Offer AI Solutions To Sports Federations", "url": "https://inc42.com/buzz/stupa-sports-analytics-nets-inr-28-cr-to-offer-ai-solutions-to-sports-federations/"}, {"title": "Stupa Sports Analytics raises Rs 28 cr to fuel innovation, expand globally", "url": "https://www.prokerala.com/news/articles/a1531284.html"}, {"title": "Stupa wins BWF approval for badminton review system", "url": "https://itbrief.in/story/stupa-wins-bwf-approval-for-badminton-review-system"}, {"title": "Badminton, Padel, Table Tennis to use AI driven IRS after BWF Approval: WTT, India Open set for New Review Procedure - mykhel", "url": "https://www.mykhel.com/more-sports/badminton-padel-table-tennis-to-use-ai-driven-irs-after-bwf-approval-wtt-india-open-set-for-new-420431.html"}]

Based on the search results, here's what I found about the sports startups and developments you mentioned:

Game Theory Sports Startup

Game Theory is a seed company based in Bengaluru, founded in 2019 by Sudeep Kulkarni. The company converts facilities into smart courts and gamifies the experience of playing sports by using smart courts to track scores, steps, heatmaps, and strokes.

Game Theory has raised \$3.13M in funding from investors like Techstars, Zerodha and Rainmatter. Notably, the startup closed its 'Masters' funding round with investments from Indian sporting legends including Padma Bhushan awardee and chief national badminton coach Pullela Gopichand, along with former professional squash player Saurav Ghosal and renowned table tennis player Sharath Kamal.

Stupa Sports Analytics

Stupa Sports Analytics, a sports technology company, has secured Rs 280 million in a pre-Series A funding round led by Centre Court Capital and PeerCapital. Founded in 2020 by Megha Gambhir and Deepak Malik, Stupa Sports Analytics offers AI-driven solutions like player performance analytics to sports federations to help them digitise their operations and management.

The company caters to Tier 2 sports like badminton, padel, pickleball, volleyball and rugby, among others.

Stupa's BWF Recognition

Recently, Stupa Sports has received certification from the Badminton World Federation for its Instant Review System for use in sanctioned tournaments, allowing the Gurugram-based sports technology company's AI-based line-calling and review product to be used across BWF events. BWF testing found the system achieved 99% accuracy, above the 98% threshold required for approval.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

14. user (2026-06-10T06:01:52.806Z)

```
backend\eval\__init__.py
backend\eval\annotations.py
backend\eval\classifier.py
backend\eval\metrics.py
backend\eval\training.py
backend\eval\calibration.py
backend\eval\gemini_refine.py
backend\eval\calibrate_local.py
backend\eval\calibrate_wasb.py
backend\eval\distill_local.py
backend\eval\export_wasb_segments.py
backend\eval\harness.py
backend\eval\regression.py
backend\eval\batch.py
backend\eval\eval_partial_labels.py
backend\eval\rally_seq_proto.py
```

15. user (2026-06-10T06:01:57.463Z)

Web search results for query: "SwingVision neighboring court filter adjacent court detection multiple courts badminton support 2026"

Links: [{"title": "SwingVision FAQ | Tennis & Pickleball App Support", "url": "https://swing.vision/faq"}, {"title": "badminton court detection Instance Segmentation Model by josephlu", "url": "https://universe.roboflow.com/josephlu/badminton-court-"}]

```
detection-cfgah"}],{"title":"Swing Court | AI-Powered Smart Tennis Courts","url":"https://swing.vision/swing-court"}, {"title":"SwingVision: AI Stats for Tennis & Pickleball","url":"https://swing.vision/"},"badminton-court-detection Object Detection Dataset and Pre-Trained Model by CVCW","url":"https://universe.roboflow.com/cvcw/badminton-court-detection"}, {"title":"SwingVision - ITA #WeAreCollegeTennis","url":"https://wearecollegetennis.com/swingvision/"},"SwingVision Smart Court - Complete AI system for tennis - Tennisnerd.net","url":"https://www.tennisnerd.net/news/swingvision-smart-court-complete-ai-system-for-tennis/45328"}]
```

Based on the search results, I found information about SwingVision, but the results don't contain specific details about a "neighboring court filter" or "adjacent court detection" feature for multiple courts or badminton support in 2026.

Here's what I found about SwingVision:

SwingVision is the #1 tennis & pickleball app for automated scoring, stats, highlights & line calling. The platform provides real-time shot tracking, video analysis and live streaming, all from just a single iPhone or iPad.

The search results mention that SwingVision allows users to control all recordings and streams from one location with no need to manually set up each court, suggesting it has multi-court capabilities. However, the specific features you're asking about regarding neighboring court filtering, adjacent court detection, or badminton support in 2026 are not detailed in these search results.

The results indicate that SwingVision primarily focuses on tennis and pickleball rather than badminton. If you need more specific information about these features, I'd recommend contacting SwingVision directly or checking their official documentation.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

16. assistant (2026-06-10T06:03:13.093Z)

CRITIQUE OF PMF RESEARCH REPORT (verified 2026-06-10 against repo docs + live web)

VERIFICATION OF LOAD-BEARING CLAIMS (5 checked, all substantively hold)

- Machaxi \$1.5M (Rainmatter-led, Prakash Padukone participating, June 2025, 1,000+ centres planned with Padukone School): CONFIRMED (entrepreneur.com/india, Outlook Business, Inc42-adjacent coverage).
- VISIST.AI (IISc/FSID-incubated, 70+ Bengaluru academies, Padukone School partnership, technique/posture assessment): CONFIRMED (ThePrint, Devdiscourse). Report's caveat that it does skill assessment, not rally segmentation, is honest and correct.
- SwingVision Wefunder financials (FY2024 revenue \$2,746,593, +12% YoY from \$2,455,089; 20k+ paying subscribers; \$4M+ ARR claim): CONFIRMED (kingscrowd.com/swingvision-on-wefunder-2025).
- PB Vision API at \$8/hr HD video (\$12/hr 4K): CONFIRMED (help.pb.vision API Partner Guide).
- Gemini 2.5 Flash \$0.30/1M input, Gemini 3 Flash \$0.50/1M: CONFIRMED as live prices (ai.google.dev/gemini-api/docs/pricing mirrors).
- Game Theory (Bengaluru, Gopichand/Bopanna/Sharath Kamal + Rainmatter/Zerodha, ~\$3.13M total, smart courts): CONFIRMED. Stupa INR 28 Cr (~\$3.3M) pre-Series A + BWF approval for its Instant Review System: CONFIRMED.
- Internal line citations all check out against the repo: PMF_MARKET_RESEARCH.md:46-48 (pricing anchors), :139 (SwingVision row incl. the 200k+ monthly users figure), :163-164 ("incumbents ignore badminton... BadPro+ is early"), :189-191 (tripod risk), :201-203 (competitive response); COMMERCIALIZATION.md:40 (C13 "Unvalidated"), :122-124 (~\$0.05/10-min input), :141-144 (whitespace framing); OWNED_MODEL_IMPLEMENTATION_PLAN.md:21 (moat = dataset+harness), :46-52 (head-before-VLM); NEXT_STEPS.md:17-18 (LOVO 0.583 vs 0.480), :20-22 (WASB TT 0.909 / pickleball 0.308), :23-25 (multi-court field notes). Confirmed the internal PMF competitive table (PMF_MARKET_RESEARCH.md:132-141) contains zero Indian competitors ? the report's core falsification is fair.
- backend/eval/eval_partial_labels.py and backend/eval/rally_seq_proto.py exist, so falsification path #2 in the verdict is actionable as stated.

FLAGS ? STALE / CHERRY-PICKED CLAIMS

- Gemini cost verdict cherry-picks the two cheapest price points and misses the price DIRECTION: the current stable flagship Gemini 3.5 Flash (May 2026) is \$1.50/1M input / \$9.00/1M output ? 5x the 2.5 Flash price the cost model rests on; trade press is explicitly calling the end of cheap Flash-tier inference. COGS still clears PB Vision's \$8/hr easily, but "cost model verified, margin wide" should carry a model-depreciation/repricing caveat (COMMERCIALIZATION.md:122-124 prices a model two generations old). The report's "Gemini 3 Flash preview at \$0.50 doesn't break the model" is the cheapest possible framing of an upward trend.
- SwingVision "slow growth / decade-long grind / +12% YoY" framing is motivated. FY2022?FY2023 roughly doubled (~\$1.0M?\$2.46M); +12% is ONE fiscal year, and the company's own \$4M+ ARR claim implies post-FY2024 re-acceleration. Concluding "this category produces niche businesses, not venture-scale ones" from n=1 company and one slow year is an overreach ? and it conveniently supports the report's de-risking narrative.
- "SwingVision raised \$6M explicitly to expand into pickleball and padel" (sportspro.com) ? NOT verified in my checks; treat as unconfirmed. Likewise Baseline Vision \$2,000-2,300, Wingfield/Clutch/Padelytics, WSC/Magnifi details, and the Decathlon INR 1,500-5,000/mo coaching-fee figure went unchecked here (lower load-bearing, but the report presents them with the same confidence as the verified items ? no confidence tiering, ironically the internal doc's own convention at PMF_MARKET_RESEARCH.md:12-15).

FLAGS ? MOTIVATED REASONING IN THE VERDICT

- The multi-court wedge is promoted from "we found no software solving it in one scan" to "the confirmed gap" / "strongest technically-grounded differentiator." It is an absence-of-evidence claim; my re-search also found no refutation, but SwingVision's "Swing Court" multi-court venue product shows incumbents instrumenting multi-court venues, and the repo's own best multi-court-robust result is LOVO F1 0.583 on n=6 (NEXT_STEPS.md:17-18) ? i.e., the "moat" is currently an unsolved problem internally too. The verdict treats a hard open problem as a defensible asset.
- Verdict whiplash: it declares the whitespace FALSIFIED, then recommends selling THROUGH Machaxi/VISIST.AI with zero evidence any Indian player buys third-party CV tech; the sole supporting analogy (PB Vision x PodPlay) is US pickleball club economics, not Indian academy economics. That pivot rescues the thesis rhetorically without new evidence.
- The table-tennis opportunity bullet is internally inconsistent: it cites Stupa (founded IN table tennis, WTT/federation distribution, \$3.3M) as proof the Indian TT market is real, then calls TT "a less contested beachhead." By the report's own logic, TT is MORE captured than badminton.
- Stupa is slightly inflated as a falsifier: its BWF-approved product is line-calling/officiating (IRS, 99% accuracy threshold testing), closer to Hawk-Eye than to highlight/analysis SaaS. The "four funded competitors" is really 2 venue/coaching plays (Machaxi, Game Theory) + 1 technique-assessment vendor (VISIST.AI) + 1 officiating/federation vendor (Stupa). The falsification of "incumbents ignore badminton" stands, but the competitive overlap with THIS product is thinner than the headline implies ? the report says this in one bullet ("SURVIVES NARROWLY") and forgets it in the risks/verdict tone.

WHAT THE REPORT MISSED

- The Padukone chokepoint cuts both ways and is bigger than stated: Padukone School of Badminton itself expanded to 75 centres in year one (target 250 by 2027) and has tie-ups with BOTH Machaxi AND VISIST.AI. Indian badminton academy distribution is concentrating into ONE celebrity network ? worse lock-up than the report describes, but also a single BD deal could unlock the entire channel. The report never names this concentration.
- Gemini video-token repricing trend (above) ? directly relevant to both the cost model and the "commoditization clock" risk, which the report frames as one-directional (models get better/cheaper) when prices are currently rising.
- No check on whether SwingVision/PB Vision-class products handle badminton's actual blocker (shuttle speed/size vs ball tracking) ? the technical transferability of the funded competitors INTO this wedge is the real threat model, and the report never assesses it.
- The internal docs' own caveat is stronger than the report admits: COMMERCIALIZATION.md:130-131 says benchmark \$/video on the target VM BEFORE pricing ? self-host costs are unmeasured, so "verified cost model" applies only to path (C) Gemini, not the owned-model endgame the repo is actually building toward.
- The report never asks whether the 2026-06-07 snapshot's miss was methodological (English-language/Western-vendor search bias) ? its own process-risk bullet says "recurring gate" but doesn't say WHY the miss happened, which is what would prevent recurrence.

BOTTOM LINE ON THE REPORT

- Core falsification (Indian competitor set absent from PMF_MARKET_RESEARCH.md §4/§5) is

accurate, well-sourced, and the report's most valuable contribution. The "narrower wedge survives" reading is defensible. The verdict's falsification-path ordering (C13 interviews ? Gemini-wrapper baseline vs LOVO ? one paid pilot) is concrete and grounded in real repo artifacts. Discount the verdict's confidence on: the multi-court moat (unproven, both in market and in repo at F1 0.583), the partner-channel rescue (evidence-free), the TT beachhead (contradicts its own Stupa evidence), and the "wide margin forever" cost framing (Gemini price direction is up).